

4

UNIT 2 :
Algebra**Linear Inequations**

(In one variable)

4.1 Introduction :

If x and y are two quantities; then both of these quantities will satisfy any one of the following four conditions (relations) :

i.e. either (i) $x > y$ (ii) $x \geq y$ (iii) $x < y$ or (iv) $x \leq y$

Each of the four conditions, given above, is an **inequation**.

In the same way, each of the following also represents an inequation :

$x < 8$, $x \geq 5$, $-x + 4 \leq 3$, $x + 8 > 4$, etc.

4.2 Linear Inequations In One Variable :

If a , b and c are real numbers, then each of the following is called a **linear inequation** in one variable :

- (i) $ax + b > c$. Read as : $ax + b$ is *greater than* c .
- (ii) $ax + b < c$. Read as : $ax + b$ is *less than* c .
- (iii) $ax + b \geq c$. Read as : $ax + b$ is *greater than or equal to* c .
- (iv) $ax + b \leq c$. Read as : $ax + b$ is *less than or equal to* c .

In an inequation, the signs ' $>$ ', ' $<$ ', ' $\geq$ ' and ' $\leq$ ' are called signs of inequality.

4.3 Solving a Linear Inequation Algebraically :

To solve a given linear inequation means to find the value or values of the variable used in it.

Thus; (i) to solve the inequation $3x + 5 > 8$ means to find the variable x .

(ii) to solve the inequation $8 - 5y \leq 3$ means to find the variable y and so on

The following working rules must be adopted for solving a given linear inequation :

Rule 1 : On transferring a positive term from one side of an inequation to its other side, the sign of the term becomes negative.

e.g. $2x + 3 > 7 \Rightarrow 2x > 7 - 3$, $5x + 4 \leq 15 \Rightarrow 5x \leq 15 - 4$,
 $23 \geq 4x + 15 \Rightarrow 23 - 15 \geq 4x$ and so on.

Rule 2 : On transferring a negative term from one side of an inequation to its other side, the sign of the term becomes positive.

e.g. $2x - 3 > 7 \Rightarrow 2x > 7 + 3$, $5x - 4 \leq 15 \Rightarrow 5x \leq 15 + 4$,
 $23 \geq 4x - 15 \Rightarrow 23 + 15 \geq 4x$ and so on.

Rule 3 : If each term of an inequation be multiplied or divided by the same positive number, the sign of inequality remains the same.

That is, if p is positive

$$(i) \quad x < y \Rightarrow px < py \quad \text{and} \quad \left(\frac{x}{p}\right) < \left(\frac{y}{p}\right),$$

$$(ii) \quad x > y \Rightarrow px > py \quad \text{and} \quad \left(\frac{x}{p}\right) > \left(\frac{y}{p}\right),$$

$$(iii) \quad x \leq y \Rightarrow px \leq py \quad \text{and} \quad \left(\frac{x}{p}\right) \leq \left(\frac{y}{p}\right)$$

$$\text{and, (iv) } x \geq y \Rightarrow px \geq py \quad \text{and} \quad \left(\frac{x}{p}\right) \geq \left(\frac{y}{p}\right).$$

$$\text{Thus, } x \leq 6 \Rightarrow 4x \leq 4 \times 6,$$

$$x \geq 5 \Rightarrow 3x \geq 3 \times 5,$$

$$x \leq 2 \Rightarrow \frac{x}{10} \leq \frac{2}{10} \quad \text{and so on.}$$

Rule 4 : If each term of an inequation be multiplied or divided by the same negative number, the sign of inequality reverses.

That is, if p is negative

$$(i) \quad x < y \Rightarrow px > py \quad \text{and} \quad \left(\frac{x}{p}\right) > \left(\frac{y}{p}\right)$$

$$(ii) \quad x \geq y \Rightarrow px \leq py \quad \text{and} \quad \left(\frac{x}{p}\right) \leq \left(\frac{y}{p}\right)$$

$$\text{Thus, } x \leq 6 \Rightarrow -4x \geq -4 \times 6,$$

$$x > 5 \Rightarrow -3x < -3 \times 5$$

$$x \geq y \Rightarrow \frac{x}{-2} \leq \frac{y}{-2} \quad \text{and so on.}$$

Rule 5 : If sign of each term on both the sides of an inequation is changed, the sign of inequality gets reversed.

$$\text{i.e. (i) } -x > 5 \Leftrightarrow x < -5 \quad (ii) \quad 3y \leq 15 \Leftrightarrow -3y \geq -15$$

$$(iii) \quad -2y < -7 \Leftrightarrow 2y > 7 \quad \text{and so on.}$$

Rule 6 : If both the sides of an inequation are positive or both are negative, then on taking their reciprocals, the sign of inequality reverses.

i.e. if x and y both are either positive or both are negative, then

$$(i) \quad x > y \Leftrightarrow \frac{1}{x} < \frac{1}{y} \quad (ii) \quad x \leq y \Leftrightarrow \frac{1}{x} \geq \frac{1}{y}$$

$$(iii) \quad x \geq y \Leftrightarrow \frac{1}{x} \leq \frac{1}{y} \quad \text{and so on.}$$

4.4 Replacement Set and Solution Set :

The set, from which the value of the variable x is to be chosen, is called **replacement set** and its subset, whose elements satisfy the given inequation, is called **solution set**.

e.g. Let the given inequation be $x < 3$, if :

- (i) the replacement set = N , the set of natural numbers;
the solution set = $\{1, 2\}$
- (ii) the replacement set = W , the set of whole numbers;
the solution set = $\{0, 1, 2\}$
- (iii) the replacement set = Z or I , the set of integers;
the solution set = $\{\dots\dots\dots, -2, -1, 0, 1, 2\}$

But, if the replacement set is the set of real numbers, the solution set can only be described in set-builder form, i.e. $\{x : x \in R \text{ and } x < 3\}$.

1 If the replacement set is the set of natural numbers (N), find the solution set of : (i) $3x + 4 < 16$ (ii) $8 - x \leq 4x - 2$.

Solution :

$$\begin{aligned}
 \text{(i) } 3x + 4 < 16 &\Rightarrow 3x < 16 - 4 && \text{[Using rule 1]} \\
 &\Rightarrow 3x < 12 \\
 &\Rightarrow \frac{3x}{3} < \frac{12}{3} && \text{[Using rule 3]} \\
 \text{i.e. } &x < 4
 \end{aligned}$$

Since, the replacement set = N (set of natural numbers)

$\therefore$ Solution set = $\{1, 2, 3\}$

Ans.

$$\begin{aligned}
 \text{(ii) } 8 - x \leq 4x - 2 &\Rightarrow -x - 4x \leq -2 - 8 && \text{[Using rule 1]} \\
 &\Rightarrow -5x \leq -10 \\
 &\Rightarrow \frac{-5x}{-5} \geq \frac{-10}{-5} && \text{[Using rule 4]} \\
 \text{i.e. } &x \geq 2
 \end{aligned}$$

Since, the replacement set = N

$\therefore$ Solution set = $\{2, 3, 4, 5, 6, \dots\dots\dots\}$

Alternative method :

$$\begin{aligned}
 8 - x \leq 4x - 2 &\Rightarrow 4x - 2 \geq 8 - x && [x \leq y \text{ and } y \geq x \text{ mean the same}] \\
 &\Rightarrow 4x + x \geq 8 + 2 && \text{[Using rule 2]} \\
 &\Rightarrow 5x \geq 10 \\
 &\Rightarrow x \geq 2 \text{ and } x \in N
 \end{aligned}$$

$\therefore$ Solution set = $\{2, 3, 4, 5, 6, \dots\dots\dots\}$

Ans.

- 2** If the replacement set is the set of whole numbers (W), find the solution set of : (i) $5x + 4 \leq 24$ (ii) $4x - 2 < 2x + 10$.

Solution :

$$\begin{aligned} \text{(i) } 5x + 4 &\leq 24 &\Rightarrow & 5x \leq 24 - 4 \\ & &\Rightarrow & 5x \leq 20 \text{ and } x \leq \frac{20}{5} \text{ i.e. } x \leq 4 \end{aligned}$$

Since, the replacement set is the set of whole numbers

$$\therefore \text{Solution set} = \{0, 1, 2, 3, 4\}$$

Ans.

$$\begin{aligned} \text{(ii) } 4x - 2 &< 2x + 10 &\Rightarrow & 4x - 2x < 10 + 2 \\ & &\Rightarrow & 2x < 12 \text{ and } x < 6 \end{aligned}$$

Since, the replacement set = W (whole numbers)

$$\therefore \text{Solution set} = \{0, 1, 2, 3, 4, 5\}$$

Ans.

- 3** If the replacement set is the set of integers, (I or Z), between -6 and 8, find the solution set of : (i) $6x - 1 \geq 9 + x$ (ii) $15 - 3x > x - 3$.

Solution :

$$\begin{aligned} \text{(i) } 6x - 1 &\geq 9 + x &\Rightarrow & 6x - x \geq 9 + 1 \\ & &\Rightarrow & 5x \geq 10 \text{ and } x \geq 2 \end{aligned}$$

Since, the replacement set is the set of integers between -6 and 8.

$$\therefore \text{Solution set} = \{2, 3, 4, 5, 6, 7\}$$

Ans.

$$\begin{aligned} \text{(ii) } 15 - 3x &> x - 3 &\Rightarrow & -3x - x > -3 - 15 \\ & &\Rightarrow & -4x > -18 \\ & &\Rightarrow & \frac{-4x}{-4} < \frac{-18}{-4} & \text{[Using rule 4]} \\ & &\Rightarrow & x < 4.5 \end{aligned}$$

Since, the replacement set is the set of integers between -6 and 8

$$\therefore \text{Solution set} = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

Ans.

Alternative method :

$$\begin{aligned} 15 - 3x &> x - 3 &\Rightarrow & x - 3 < 15 - 3x \\ & &\Rightarrow & x + 3x < 15 + 3 \\ & &\Rightarrow & 4x < 18 \text{ and } x < 4.5 \end{aligned}$$

$$\therefore \text{Solution set} = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4\}$$

Ans.

- 4** If the replacement set is the set of real numbers (R), find the solution set of : (i) $5 - 3x < 11$ (ii) $8 + 3x \geq 28 - 2x$.

Solution :

$$\text{(i) } 5 - 3x < 11 \quad \Rightarrow \quad -3x < 11 - 5$$

$$\Rightarrow -3x < 6$$

$$\Rightarrow \frac{-3x}{-3} > \frac{6}{-3}$$

[Using rule 4]

$$\Rightarrow x > -2$$

Since, the replacement set is the set of real numbers (R)

$$\therefore \text{Solution set} = \{x : x > -2 \text{ and } x \in \mathbf{R}\}$$

Ans.

$$(ii) \quad 8 + 3x \geq 28 - 2x \quad \Rightarrow \quad 3x + 2x \geq 28 - 8$$

$$\Rightarrow \quad 5x \geq 20 \quad \text{and} \quad x \geq 4$$

Since, the replacement set is the set of real numbers

$$\therefore \text{Solution set} = \{x : x \geq 4 \text{ and } x \in \mathbf{R}\}$$

Ans.

5 Solve : $\frac{x}{2} - 5 \leq \frac{x}{3} - 4$, where x is a positive odd integer.

Solution :

$$\frac{x}{2} - 5 \leq \frac{x}{3} - 4 \quad \Rightarrow \quad \frac{x}{2} - \frac{x}{3} \leq -4 + 5$$

$$\Rightarrow \quad \frac{3x - 2x}{6} \leq 1 \quad \Rightarrow \quad x \leq 6$$

Since, x is a positive odd integer

$$\therefore \text{Solution set} = \{1, 3, 5\}$$

Ans.

6 Solve the following inequation : $2y - 3 < y + 1 \leq 4y + 7$; if :

(i) $y \in \{\text{Integers}\}$ (ii) $y \in \mathbf{R}$ (real numbers)

Solution :

$$2y - 3 < y + 1 \leq 4y + 7 \quad \Rightarrow \quad 2y - 3 < y + 1 \quad \text{and} \quad y + 1 \leq 4y + 7$$

$$\Rightarrow \quad y < 4 \quad \text{and} \quad -6 \leq 3y$$

$$\Rightarrow \quad y < 4 \quad \text{and} \quad y \geq -2$$

$$\Rightarrow \quad -2 \leq y < 4$$

(i) When $y \in \{\text{Integers}\}$

$$\therefore \text{Solution set} = \{-2, -1, 0, 1, 2, 3\}$$

Ans.

(ii) When $y \in \mathbf{R}$ (real numbers)

$$\therefore \text{Solution set} = \{y : -2 \leq y < 4 \text{ and } y \in \mathbf{R}\}$$

Ans.

EXERCISE 4(A)

1. State, true or false :

- (i) $x < -y \Rightarrow -x > y$
 (ii) $-5x \geq 15 \Rightarrow x \geq -3$
 (iii) $2x \leq -7 \Rightarrow \frac{2x}{-4} \geq \frac{-7}{-4}$
 (iv) $7 > 5 \Rightarrow \frac{1}{7} < \frac{1}{5}$

2. State, whether the following statements are true or false.

- (i) If $a < b$, then $a - c < b - c$
 (ii) If $a > b$, then $a + c > b + c$
 (iii) If $a < b$, then $ac > bc$
 (iv) If $a > b$, then $\frac{a}{c} < \frac{b}{c}$
 (v) If $a - c > b - d$; then $a + d > b + c$
 (vi) If $a < b$, and $c > 0$, then $a - c > b - c$
 where a, b, c and d are real numbers and $c \neq 0$.

3. If $x \in \mathbb{N}$, find the solution set of inequations.

- (i) $5x + 3 \leq 2x + 18$
 (ii) $3x - 2 < 19 - 4x$

4. If the replacement set is the set of whole numbers, solve :

- (i) $x + 7 \leq 11$ (ii) $3x - 1 > 8$
 (iii) $8 - x > 5$ (iv) $7 - 3x \geq -\frac{1}{2}$

(v) $x - \frac{3}{2} < \frac{3}{2} - x$ (vi) $18 \leq 3x - 2$

5. Solve the inequation :

$3 - 2x \geq x - 12$ given that $x \in \mathbb{N}$.

6. If $25 - 4x \leq 16$, find :

- (i) the smallest value of x , when x is a real number,
 (ii) the smallest value of x , when x is an integer.

7. If the replacement set is the set of real numbers, solve :

(i) $-4x \geq -16$ (ii) $8 - 3x \leq 20$

(iii) $5 + \frac{x}{4} > \frac{x}{5} + 9$ (iv) $\frac{x+3}{8} < \frac{x-3}{5}$

8. Find the smallest value of x for which

$5 - 2x < 5\frac{1}{2} - \frac{5}{3}x$, where x is an integer.

9. Find the largest value of x for which

$2(x - 1) \leq 9 - x$ and $x \in \mathbb{W}$.

10. Solve the inequation :

$12 + 1\frac{5}{6}x \leq 5 + 3x$ and $x \in \mathbb{R}$.

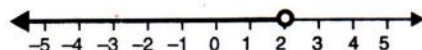
11. Given $x \in \{\text{integers}\}$, find the solution set of : $-5 \leq 2x - 3 < x + 2$.12. Given $x \in \{\text{whole numbers}\}$, find the solution set of : $-1 \leq 3 + 4x < 23$.**4.5 Representation of the solution on the number line :****A real number line can be used to represent the solution set of an inequation.**

The convention is that $\circ$ (a hollow circle) marks the end of a range with a *strict inequality* (i.e. $<$ or $>$) and $\bullet$ (a darkened circle) marks the end of a range involving an equality as well (i.e. $\leq$ or $\geq$).

For Example :

The adjacent figure shows

$x < 2$ and $x \in \mathbb{R}$.

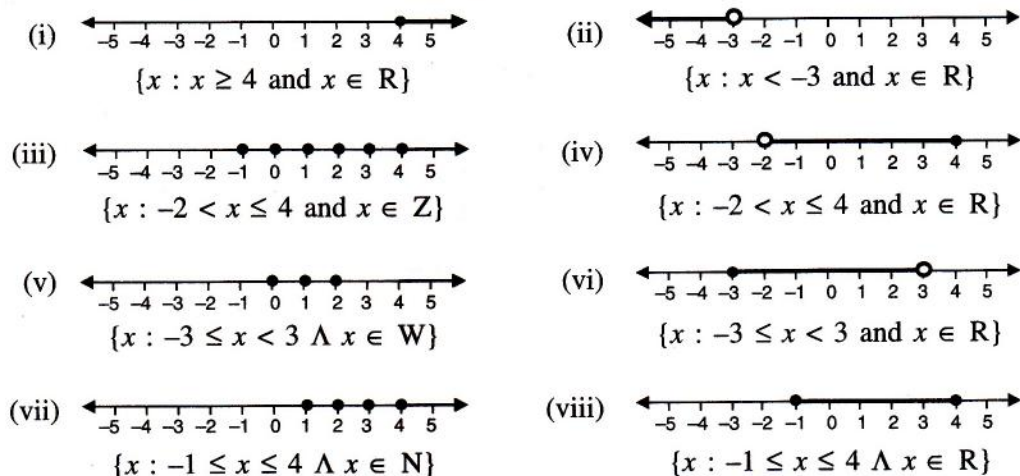


The number 2 is encircled and the circle is not darkened to show that 2 is not included in the graph.

If 2 is also included i.e. $x \leq 2$, then the circle will be darkened and the graph will be as shown alongside:



Similarly, study each of the following carefully :



[Note: The symbol ' $\wedge$ ' stands for 'and'].

- 7** Given that $x \in \mathbb{R}$, solve the following inequality and graph the solution on the number line : $-1 \leq 3 + 4x < 23$ [2006]

Solution :

Given : $-1 \leq 3 + 4x < 23; x \in \mathbb{R}$

$$\Rightarrow -1 \leq 3 + 4x \text{ and } 3 + 4x < 23$$

$$\Rightarrow -4 \leq 4x \text{ and } 4x < 20$$

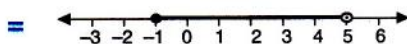
$$\Rightarrow -1 \leq x \text{ and } x < 5$$

$$\Rightarrow -1 \leq x < 5; x \in \mathbb{R}$$

$$\therefore \text{Solution} = \{-1 \leq x < 5 : x \in \mathbb{R}\}$$

Ans.

Solution on the number line is :



Ans.

- 8** Simplify : $-\frac{1}{3} \leq \frac{x}{2} - 1\frac{1}{3} < \frac{1}{6} : x \in \mathbb{R}$.

Graph the values of x on the real number line.

Solution :

The given inequation has two parts :

$$-\frac{1}{3} \leq \frac{x}{2} - 1\frac{1}{3} \quad \text{and} \quad \frac{x}{2} - 1\frac{1}{3} < \frac{1}{6}$$

$$\Rightarrow -\frac{1}{3} + 1\frac{1}{3} \leq \frac{x}{2} \quad \Rightarrow \quad \frac{x}{2} < \frac{1}{6} + \frac{4}{3}$$

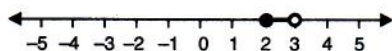
$$\Rightarrow 1 \leq \frac{x}{2} \quad \Rightarrow \quad \frac{x}{2} < \frac{9}{6}$$

$$\Rightarrow 2 \leq x \quad \Rightarrow \quad x < \frac{9}{6} \times 2$$

$$\Rightarrow x < 3$$

∴ On simplifying, the given inequation reduces to $2 \leq x < 3$ and the required graph on number line is :

Ans.



- 9** List the solution set of $50 - 3(2x - 5) < 25$, given that $x \in \mathbb{W}$.
Also, represent the solution set obtained on a number line.

Solution :

$$50 - 3(2x - 5) < 25 \Rightarrow 50 - 6x + 15 < 25$$

$$\Rightarrow -6x < 25 - 65$$

$$\Rightarrow -6x < -40$$

$$\Rightarrow \frac{-6x}{-6} > \frac{-40}{-6}$$

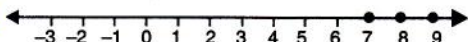
Division by a negative number
reverses the sign of inequality.

$$\Rightarrow x > 6\frac{2}{3}$$

Required solution set = $\{7, 8, 9, \dots\}$

Ans.

And, the required number line is :



Ans.

4.6 Combining Inequations :

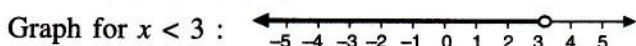
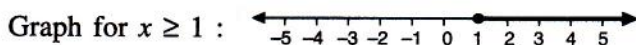
- 10** Solve and graph the solution set of $3x + 6 \geq 9$ and $-5x > -15$; where $x \in \mathbb{R}$.

Solution :

$$3x + 6 \geq 9 \Rightarrow 3x \geq 9 - 6$$

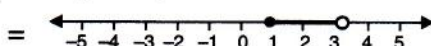
$$\Rightarrow 3x \geq 3 \Rightarrow x \geq 1$$

$$\text{And, } -5x > -15 \Rightarrow \frac{-5x}{-5} < \frac{-15}{-5} \Rightarrow x < 3$$



∴ Graph of solution set of $x \geq 1$ and $x < 3$

= Graph of points common to both $x \geq 1$ and $x < 3$



Ans.

- 11** Solve and graph the solution set of $-2 < 2x - 6$ or $-2x + 5 \geq 13$; where $x \in \mathbb{R}$.

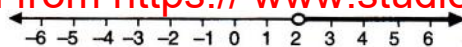
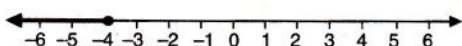
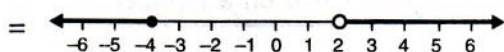
Solution :

$$-2 < 2x - 6 \Rightarrow 2x - 6 > -2$$

$$\Rightarrow 2x > 4 \Rightarrow x > 2$$

$$-2x + 5 \geq 13 \Rightarrow -2x \geq 8$$

$$\Rightarrow x \leq -4$$

Graph for $x > 2$:Graph for $x \leq -4$: $\therefore$ Graph of solution set of $x > 2$ or $x \leq -4$ = Graph of points which belong to $x > 2$ or $x \leq -4$ or both

Ans.

12Given : $P = \{x : 5 < 2x - 1 \leq 11, x \in \mathbb{R}\}$ $Q = \{x : -1 \leq 3 + 4x < 23, x \in \mathbb{I}\}$ where $\mathbb{R} = \{\text{real numbers}\}$ and $\mathbb{I} = \{\text{integers}\}$.Represent P and Q on two different number lines. Write down the elements of $P \cap Q$.**Solution :****For P :** $5 < 2x - 1 \leq 11$, where $x \in \mathbb{R}$

$$\Rightarrow 5 < 2x - 1 \quad \text{and} \quad 2x - 1 \leq 11$$

$$\Rightarrow 3 < x \quad \text{and} \quad x \leq 6$$

$$\text{i.e. } 3 < x \leq 6; x \in \mathbb{R}$$

$$\therefore P = \text{Number line from -4 to 8 with an open circle at 3 and a closed circle at 6, shaded between them.}$$

For Q : $-1 \leq 3 + 4x < 23$, $x \in \mathbb{I}$

$$\Rightarrow -1 \leq 3 + 4x \quad \text{and} \quad 3 + 4x < 23$$

$$\Rightarrow -1 \leq x \quad \text{and} \quad x < 5$$

$$\text{i.e. } -1 \leq x < 5; \text{ where } x \in \mathbb{I}$$

$$\therefore Q = \text{Number line from -4 to 8 with closed circles at -1 and 4, shaded between them.}$$

Hence, $P \cap Q = \{\text{elements common to both } P \text{ and } Q\} = \{4\}$.

Ans.

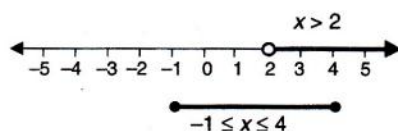
13Write down the range of values of x ($x \in \mathbb{R}$) for which both the inequations $x > 2$ and $-1 \leq x \leq 4$ are true.**Solution :**

Both the given inequations are true in the range; where their graphs on the real number lines overlap.

The graphs for the given inequalities are drawn alongside :

It is clear from both the graphs that their common range is $2 < x \leq 4$. $\therefore$ Required range is $2 < x \leq 4$

Ans.

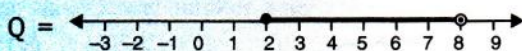
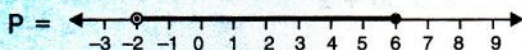
For any two solution sets A and B :(i) A and B = Intersection of sets A and B = Set of elements common to set A and to set B

$$= A \cap B$$

(ii) A or B = Union of sets A and B = Set of elements which belong either set A and to set B

$$= A \cup B$$

- 14** The diagram, given below, represents two inequations P and Q on real number lines :



- (i) Write down P and Q in set builder notation.
 (ii) Represent each of the following sets on different number lines :

(a) $P \cup Q$

(b) $P \cap Q$

(c) $P - Q$

(d) $Q - P$

(e) $P \cap Q'$

(f) $P' \cap Q$

Solution :

(i) $P = \{x : -2 < x \leq 6 \text{ and } x \in \mathbb{R}\}$ and,

$Q = \{x : 2 \leq x < 8 \text{ and } x \in \mathbb{R}\}$

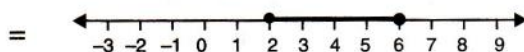
Ans.

(ii) (a) $P \cup Q$ = Numbers which belong to P or to Q or to both



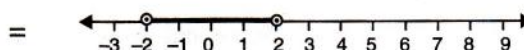
Ans.

(b) $P \cap Q$ = Numbers common to both P and Q



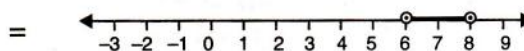
Ans.

(c) $P - Q$ = Numbers which belong to P but do not belong to Q



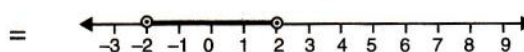
Ans.

(d) $Q - P$ = Numbers which belong to Q but do not belong to P



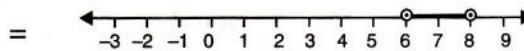
Ans.

(e) $P \cap Q'$ = Numbers which belong to P but do not belong to Q = $P - Q$



Ans.

(f) $P' \cap Q$ = Numbers which do not belong to P but belong to Q = $Q - P$



Ans.

- 15** Find three smallest consecutive whole numbers such that the difference between one-fourth of the largest and one-fifth of the smallest is at least 3.

Solution :

Let the required whole numbers be x , $x + 1$ and $x + 2$.

According to the given statement :

$$\frac{x+2}{4} - \frac{x}{5} \geq 3 \Rightarrow \frac{5x+10-4x}{20} \geq 3$$

$$\Rightarrow x + 10 \geq 60 \text{ i.e., } x \geq 50$$

Since, the smallest value of $x = 50$ that satisfies the inequation $x \geq 50$.

∴ Required smallest consecutive whole numbers are :

$$= x, x + 1 \text{ and } x + 2$$

$$= 50, 50 + 1 \text{ and } 50 + 2$$

$$= 50, 51 \text{ and } 52$$

Ans.

EXERCISE 4(B)

1. Represent the following inequalities on real number lines :

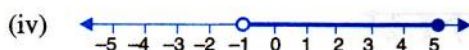
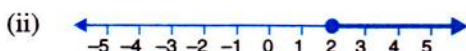
(i) $2x - 1 < 5$ (ii) $3x + 1 \geq -5$

(iii) $2(2x - 3) \leq 6$ (iv) $-4 < x < 4$

(v) $-2 \leq x < 5$ (vi) $8 \geq x > -3$

(vii) $-5 < x \leq -1$

2. For each graph given alongside, write an inequation taking x as the variable :



3. For the following inequations, graph the solution set on the real number line :

(i) $-4 \leq 3x - 1 < 8$

(ii) $x - 1 < 3 - x \leq 5$

4. Represent the solution of each of the following inequations on the real number line:

(i) $4x - 1 > x + 11$

(ii) $7 - x \leq 2 - 6x$

(iii) $x + 3 \leq 2x + 9$

(iv) $2 - 3x > 7 - 5x$

(v) $1 + x \geq 5x - 11$

(vi) $\frac{2x+5}{3} > 3x - 3$

5. $x \in \{\text{real numbers}\}$ and $-1 < 3 - 2x \leq 7$, evaluate x and represent it on a number line.

6. List the elements of the solution set of the inequation $-3 < x - 2 \leq 9 - 2x$; $x \in \mathbb{N}$.

7. Find the range of values of x which satisfies

$$-2\frac{2}{3} \leq x + \frac{1}{3} < 3\frac{1}{3}, x \in \mathbb{R}.$$

Graph these values of x on the number line.

[2007]

8. Find the values of x , which satisfy the inequation:

$$-2 \leq \frac{1}{2} - \frac{2x}{3} \leq 1\frac{5}{6}, x \in \mathbb{N}.$$

Graph the solution on the number line.

9. Given $x \in \{\text{real numbers}\}$, find the range of values of x for which $-5 \leq 2x - 3 < x + 2$ and represent it on a real number line.

10. If $5x - 3 \leq 5 + 3x \leq 4x + 2$, express it as $a \leq x \leq b$ and then state the values of a and b .

11. Solve the following inequation and graph the solution set on the number line :

$$2x - 3 < x + 2 \leq 3x + 5; x \in \mathbb{R}.$$

12. Solve and graph the solution set of :

(i) $2x - 9 < 7$ and $3x + 9 \leq 25$; $x \in \mathbb{R}$.

(ii) $2x - 9 \leq 7$ and $3x + 9 > 25$; $x \in \mathbb{I}$.

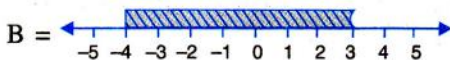
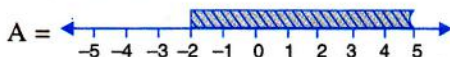
(iii) $x + 5 \geq 4(x - 1)$ and $3 - 2x < -7$; $x \in \mathbb{R}$.

13. Solve and graph the solution set of :

(i) $3x - 2 > 19$ or $3 - 2x \geq -7$; $x \in \mathbb{R}$.

(ii) $5 > p - 1 > 2$ or $7 \leq 2p - 1 \leq 17$; $p \in \mathbb{R}$.

14. The diagram represents two inequations A and B on real number lines :



(i) Write down A and B in set builder notation.

(ii) Represent $A \cap B$ and $A \cap B'$ on two different number lines.

15. Use real number line to find the range of values of x for which :

(i) $x > 3$ and $0 < x < 6$.

(ii) $x < 0$ and $-3 \leq x < 1$.

(iii) $-1 < x \leq 6$ and $-2 \leq x \leq 3$

16. Illustrate the set $\{x : -3 \leq x < 0 \text{ or } x > 2; x \in \mathbb{R}\}$ on a real number line.

17. Given $A = \{x : -1 < x \leq 5, x \in \mathbb{R}\}$ and $B = \{x : -4 \leq x < 3, x \in \mathbb{R}\}$

Represent on different number lines :

(i) $A \cap B$ (ii) $A' \cap B$ (iii) $A - B$

18. P is the solution set of $7x - 2 > 4x + 1$ and Q is the solution set of $9x - 45 \geq 5(x - 5)$; where $x \in \mathbb{R}$. Represent :

(i) $P \cap Q$

(ii) $P - Q$

(iii) $P \cap Q'$ on different number lines.

19. If $P = \{x : 7x - 4 > 5x + 2, x \in \mathbb{R}\}$ and $Q = \{x : x - 19 \geq 1 - 3x, x \in \mathbb{R}\}$; find the range of set $P \cap Q$ and represent it on a number line.

20. Find the range of values of x , which satisfy:

$$-\frac{1}{3} \leq \frac{x}{2} + 1\frac{2}{3} < 5\frac{1}{6}$$

Graph, in each of the following cases, the values of x on the different real number lines:

(i) $x \in \mathbb{W}$ (ii) $x \in \mathbb{Z}$ (iii) $x \in \mathbb{R}$.

21. Given : $A = \{x : -8 < 5x + 2 \leq 17, x \in \mathbb{I}\}$

$$B = \{x : -2 \leq 7 + 3x < 17, x \in \mathbb{R}\}$$

Where $\mathbb{R} = \{\text{real numbers}\}$ and $\mathbb{I} = \{\text{integers}\}$.

Represent A and B on two different number lines. Write down the elements of $A \cap B$.

22. Solve the following inequation and represent the solution set on the number line $2x - 5 \leq 5x + 4 < 11$, where $x \in \mathbb{I}$. [2011]

23. Given that $x \in \mathbb{I}$, solve the inequation and graph the solution on the number line :

$$3 \geq \frac{x-4}{2} + \frac{x}{3} \geq 2. \quad [2004]$$

24. Given :

$$A = \{x : 11x - 5 > 7x + 3, x \in \mathbb{R}\} \text{ and}$$

$$B = \{x : 18x - 9 \geq 15 + 12x, x \in \mathbb{R}\}.$$

Find the range of set $A \cap B$ and represent it on a number line [2005]

25. Find the set of values of x , satisfying :

$$7x + 3 \geq 3x - 5 \text{ and } \frac{x}{4} - 5 \leq \frac{5}{4} - x,$$

where $x \in \mathbb{N}$.

26. Solve :

$$(i) \frac{x}{2} + 5 \leq \frac{x}{3} + 6, \text{ where } x \text{ is a positive odd integer}$$

$$(ii) \frac{2x+3}{3} \geq \frac{3x-1}{4}, \text{ where } x \text{ is a positive even integer}$$

27. Solve the inequation :

$$-2\frac{1}{2} + 2x \leq \frac{4x}{5} \leq \frac{4}{3} + 2x, x \in \mathbb{W}.$$

Graph the solution set on the number line.

28. Find three consecutive largest positive integers such that the sum of one-third of first, one-fourth of second and one-fifth of third is at most 20.

29. Solve the given inequation and graph the solution on the number line.

$$2y - 3 < y + 1 \leq 4y + 7, y \in \mathbb{R} \quad [2008]$$

30. Solve the inequation :

$$3z - 5 \leq z + 3 < 5z - 9; z \in \mathbb{R}.$$

Graph the solution set on the number line.

31. Solve the following inequation and represent the solution set on the number line.

$$-3 < -\frac{1}{2} - \frac{2x}{3} \leq \frac{5}{6}, x \in \mathbb{R}. \quad [2010]$$

32. Solve the following inequation and represent the solution set on the number line :

$$4x - 19 < \frac{3x}{5} - 2 \leq \frac{-2}{5} + x, x \in \mathbb{R} \quad [2012]$$

33. Solve the following inequation, write the solution set and represent it on the number line:

$$-\frac{x}{3} \leq \frac{x}{2} - 1\frac{1}{3} < \frac{1}{6}, x \in \mathbb{R} \quad [2013]$$

34. Find the values of x , which satisfy the inequation

$$-2\frac{5}{6} < \frac{1}{2} - \frac{2x}{3} \leq 2, x \in \mathbb{W}. \text{ Graph the solution}$$

set on the number line. [2014]

35. Solve the following inequation and write the solution set :

$$13x - 5 < 15x + 4 < 7x + 12, x \in \mathbb{R}$$

Represent the solution on a real number line.

[2015]

4.7 Important :

1. If the product of two numbers is less than 0 (zero) i.e. product is negative; one of the two numbers is positive and the other is negative.
2. If the product of two numbers is greater than 0 (zero) i.e. positive; both the numbers are positive or both are negative.

1. For $(x - 3)(x + 5) < 0$

If $(x - 3)$ is positive, then $x + 5$ is negative

and if $x - 3$ is negative, then $x + 5$ is positive

Case 1 : When $(x - 3)$ is positive and $(x + 5)$ is negative

i.e. $x - 3 > 0$ and $x + 5 < 0$

$\Rightarrow x > 3$ and $x < -5$

It is not possible, as number greater than 3 cannot be smaller than -5 .

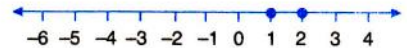
Case 2 : When $(x - 3)$ is negative and $(x + 5)$ is positive

i.e. $x - 3 < 0$ and $x + 5 > 0$

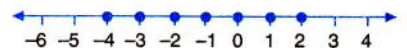
$\Rightarrow x < 3$ and $x > -5$

$\Rightarrow -5 < x < 3$, which is possible.

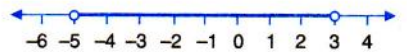
If $x \in \mathbb{N}$, then : $-5 < x < 3$ on the number line is :



If $x \in \mathbb{Z}$, then : $-5 < x < 3$ on the number line is :



If $x \in \mathbb{R}$, then : $-5 < x < 3$ on the number line is :

**2. For $(x - 3)(x + 5) > 0$**

If $(x - 3)$ is negative, then $(x + 5)$ is also negative

and, if $(x - 3)$ is positive, then $(x + 5)$ is also positive

Case 1 : $x - 3$ is negative and $x + 5$ is negative

i.e. $x - 3 < 0$ and $x + 5 < 0$

$\Rightarrow x < 3$ and $x < -5$

$\Rightarrow x < -5$ as the number smaller than -5 is less than 3 also.

Case 2 : $x - 3$ is positive and $x + 5$ is positive

i.e. $x - 3 > 0$ and $x + 5 > 0$

$\Rightarrow x > 3$ and $x > -5$

$\Rightarrow x > 3$ as the number greater than 3 is also greater than -5 .

If $x \in \mathbb{R}$, then for $(x - 3)(x + 5) > 0$ the number line is :

